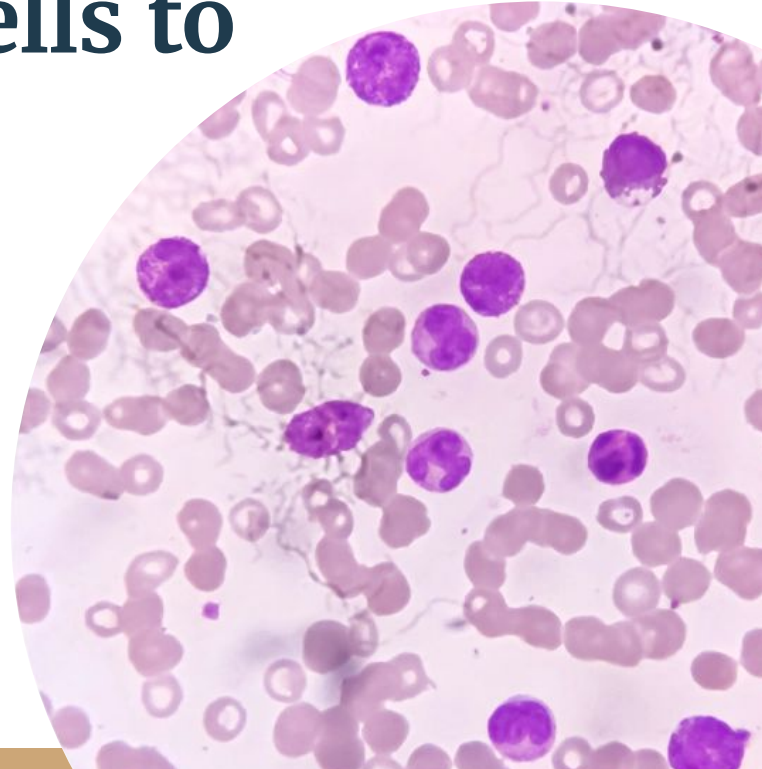


Gene expression profiling predicts sensitivity of chronic lymphocytic leukemia cells to dasatinib

Group: Ivon Sanchez, Samuel Escudero, Karim Hamed

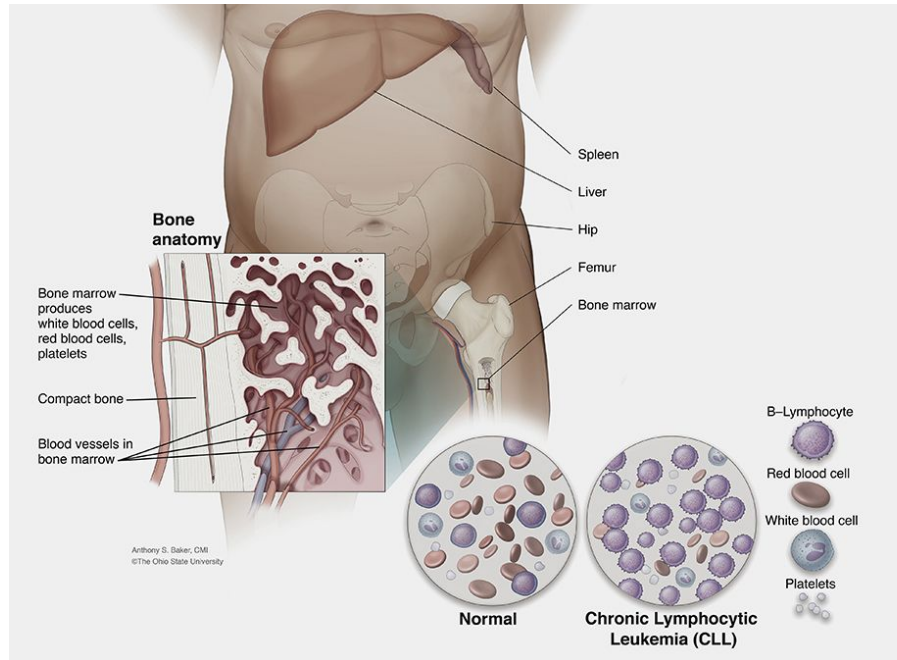
Institution: Universitat Pompeu Fabra

Date: June 4th, 2025



Background & Objective

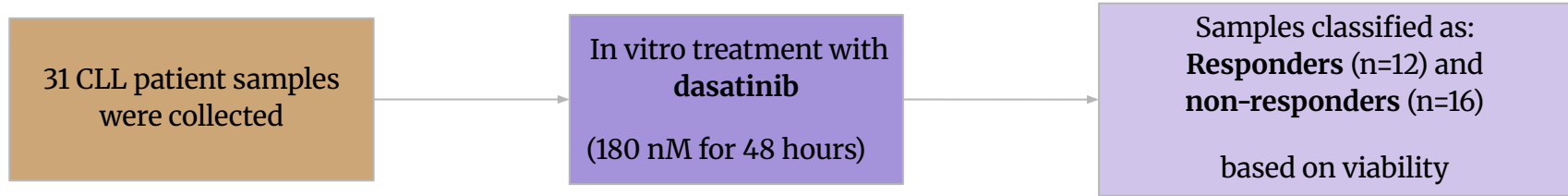
Chronic Lymphocytic Leukemia (CLL): common B-cell malignancy with a heterogeneous clinical course



Dasatinib = TKI with variable patient response

Study Question: *Can we use gene expression to predict response to dasatinib?*

Study cohort and experimental design



- Then, RNA-seq was performed on all samples

Key Findings:

- Identified **154 genes differentially expressed** between responders and nonresponders.
- Among these, **31 genes were predictive** of response in untreated samples.
- Responders exhibited higher expression of genes associated with stress signaling, glycolysis, and immune responses.

Study objective:

- Identify gene expression patterns associated with dasatinib sensitivity as potential biomarkers, Blätte et al. (2021)

Data source & Input

DATASET

- **Source:** Publicly available RNA-seq data from **CLL patient samples**, accessed via **GEO**.
- **Samples:**
 - **n = 31** total CLL samples.
 - **n = 28** used after filtering (12 Responders, 16 Non-responders to dasatinib).
- **Format:** Stored in a **SummarizedExperiment** object (**ccl**), containing:
 - **Gene expression counts** (raw + $\log_2$ CPM).
 - **Patient metadata** (response classification, sample IDs, treatment info).

Data source & Input

FILTERING

- Genes with $\log_2\text{CPM} < 0$ considered lowly expressed and filtered out.
- Final gene set: ~14,600 genes after filtering.

NORMALIZATION

- Applied TMM (Trimmed Mean of M-values) normalization using the **edgeR** package.

(Adjusts for library size differences and compositional bias.)

Data source & Input

DESIGN

- Comparison between **Responders vs. Non-responders** to dasatinib.
- Experimental design modeled with **limma-voom**, chosen for:
 - Speed and flexibility.

limma-voom pipeline overview



1. Import & QC

- Load raw counts from GEO and sample metadata.
- Check library sizes and run MDS/PCA to spot outliers.

2. Filter low-abundance Genes

- Remove genes with CPM < 1 in most samples.

3. Normalize (TMM)

- Apply calcNormFactors() to adjust for library-size differences.

4. voom transformation

- Run voom() to model mean-variance and compute weights.

5. Fit model (limma)

- Build design matrix ($\sim$ responder_status + batch) and run lmFit().

6. eBayes moderation

- Use eBayes() to shrink variances and improve p-values.

7. Identify DEGs

- Select genes with $|\log_2FC| \geq 0.58$ and $p \leq 0.01$ (FDR < 0.01).

QC and Normalization

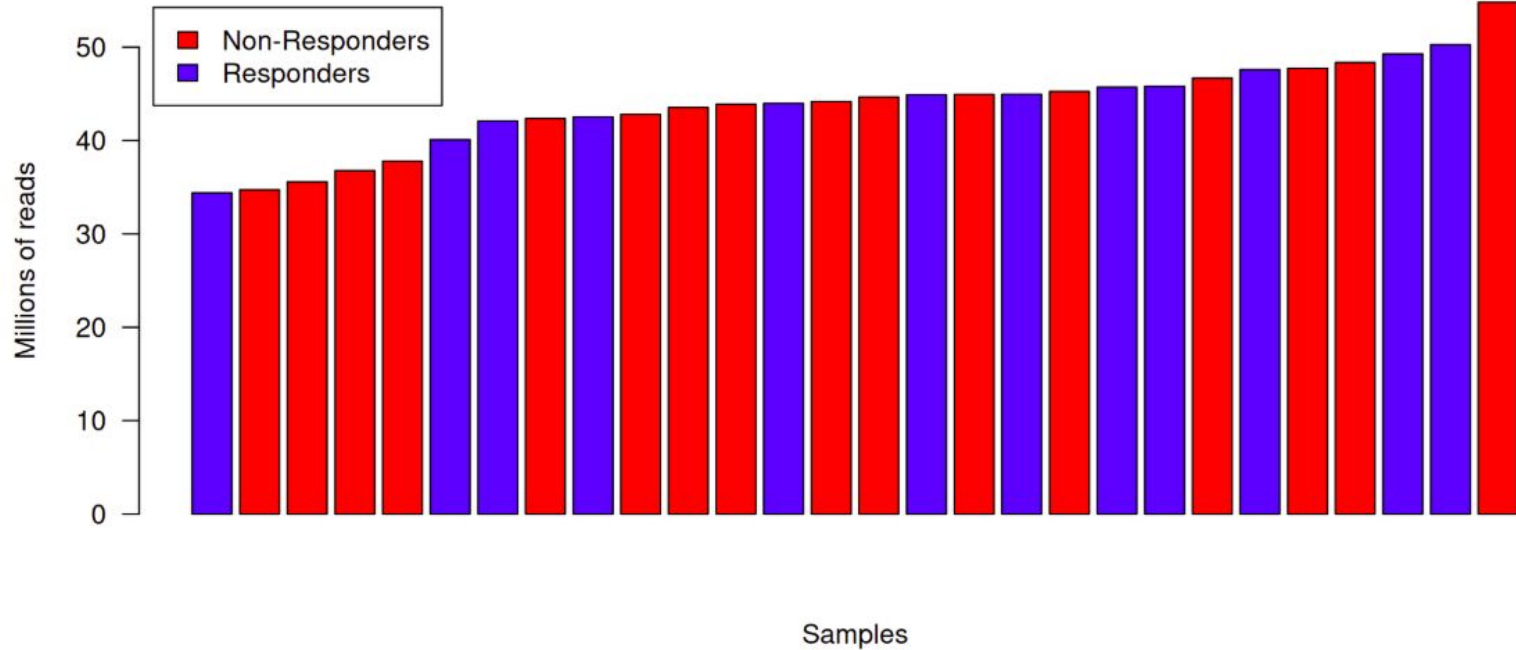


Figure 1: Barplot of sequencing depth (in millions of reads) for each sample, colored by responders and non-responders (red = non-responders, blue = responders)

Average gene expression levels ($\log_2\text{CPM}$) across all genes

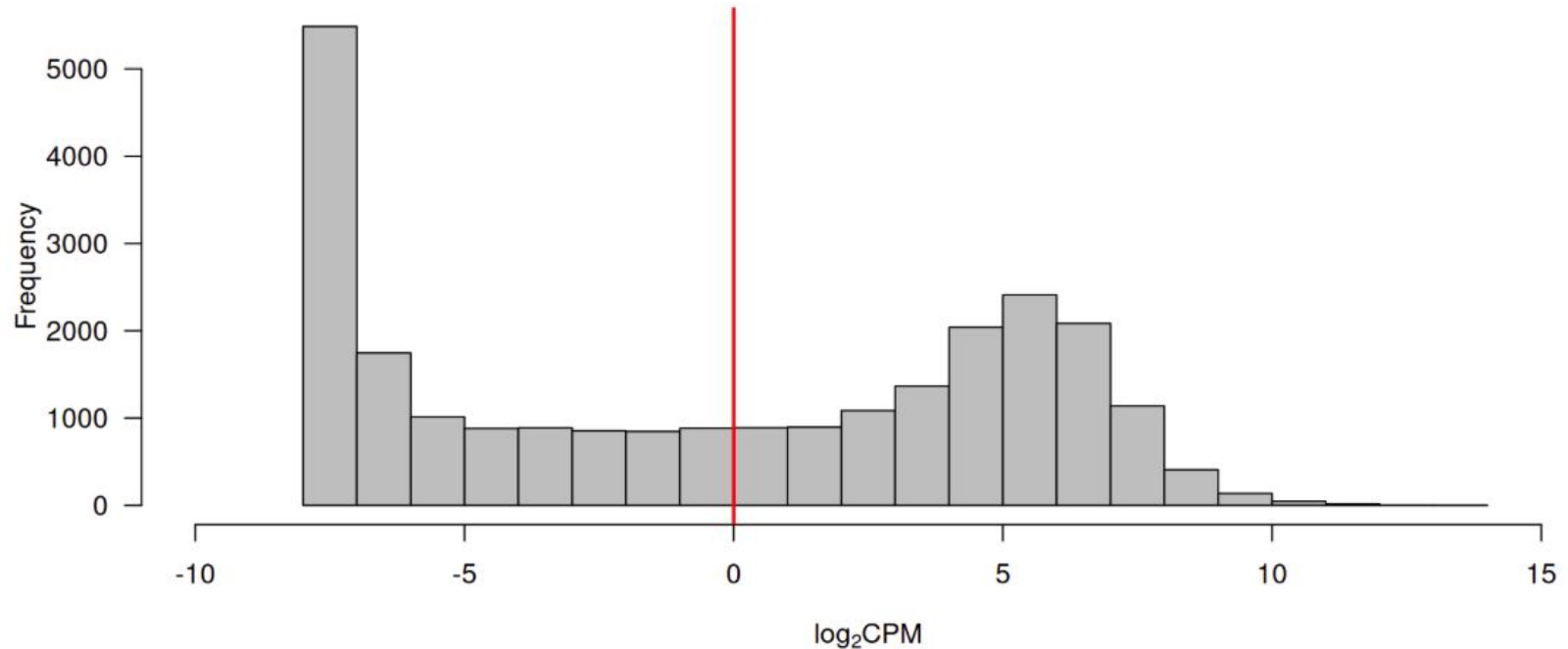


Figure 3: **Histogram of average gene expression levels ($\log_2\text{CPM}$) across all genes**

Genes with expression below 0 $\log_2\text{CPM}$ are considered lowly expressed and are filtered out.

Filtering of lowly expressed genes

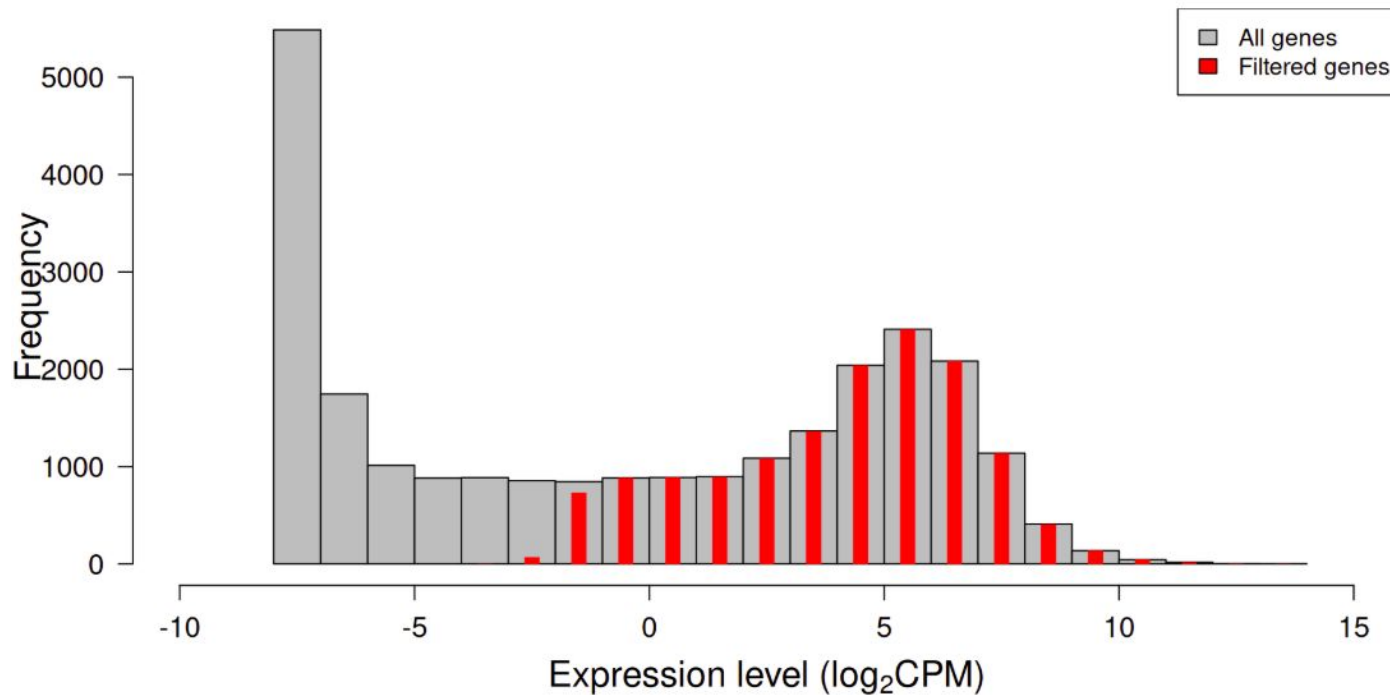


Figure 4: **Overlay of filtered genes (red) on the distribution of average gene expression levels ($\log_2\text{CPM}$) across all genes**

Sample Clustering by Dasatinib Response (MDS Plot)

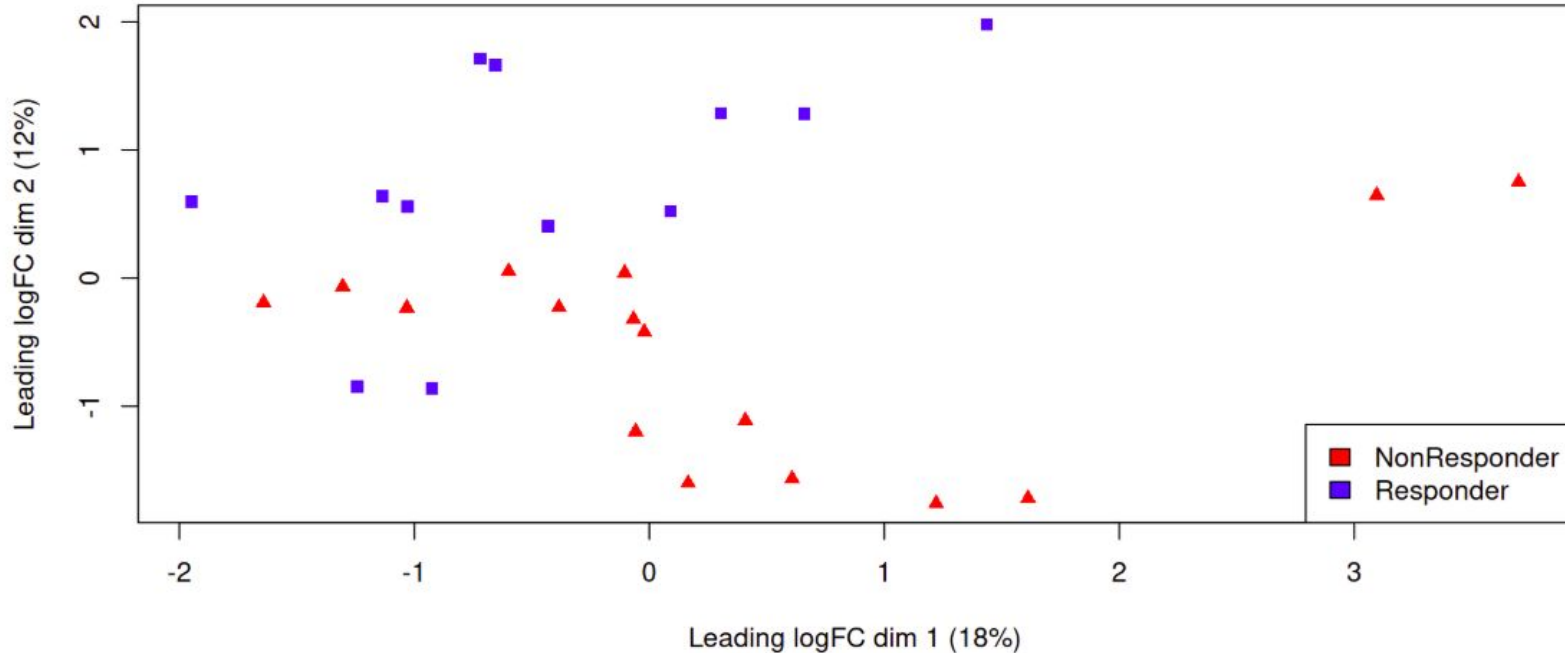


Figure 7: **MDS plot showing sample clustering by leading log-fold change distances, with color indicating dasatinib response status (red = non-responder, blue = responder)**

The clear separation along the first dimension reflects response-related expression differences.

Hierarchical Clustering of CLL Samples by Expression Profile

2.6 Experimental design

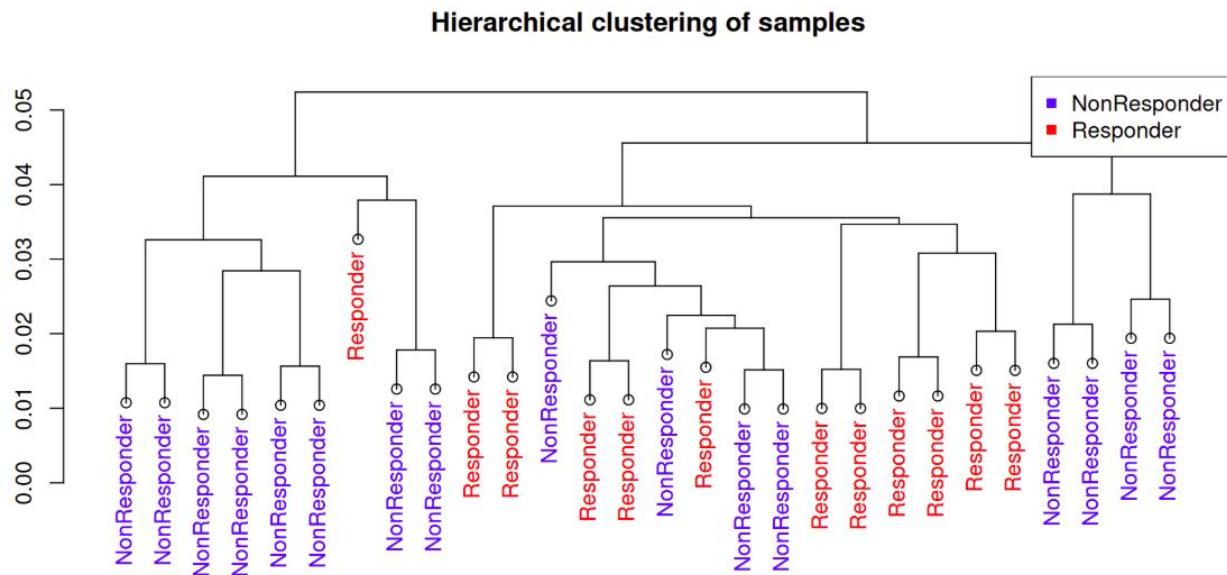


Figure 8: **Hierarchical clustering of samples based on gene expression profiles**

Samples partially group by response, but substantial overlap suggests response is not the dominant driver of variability. Response labels ('Responders'/'Non-responders') are spread across clusters, indicating no strong clustering by outcome.

Differential Expression Genes Results

- 88 DEGs identified between responders and non-responders
- Criteria:
 - $\text{FDR} \leq 0.01$ (Benjamini–Hochberg adjusted)
 - $|\log_2 \text{Fold Change}| \geq 0.58$
- Indicates statistically and biologically meaningful differences
- Suggests potential genes involved in dasatinib response

Significantly regulated expression genes

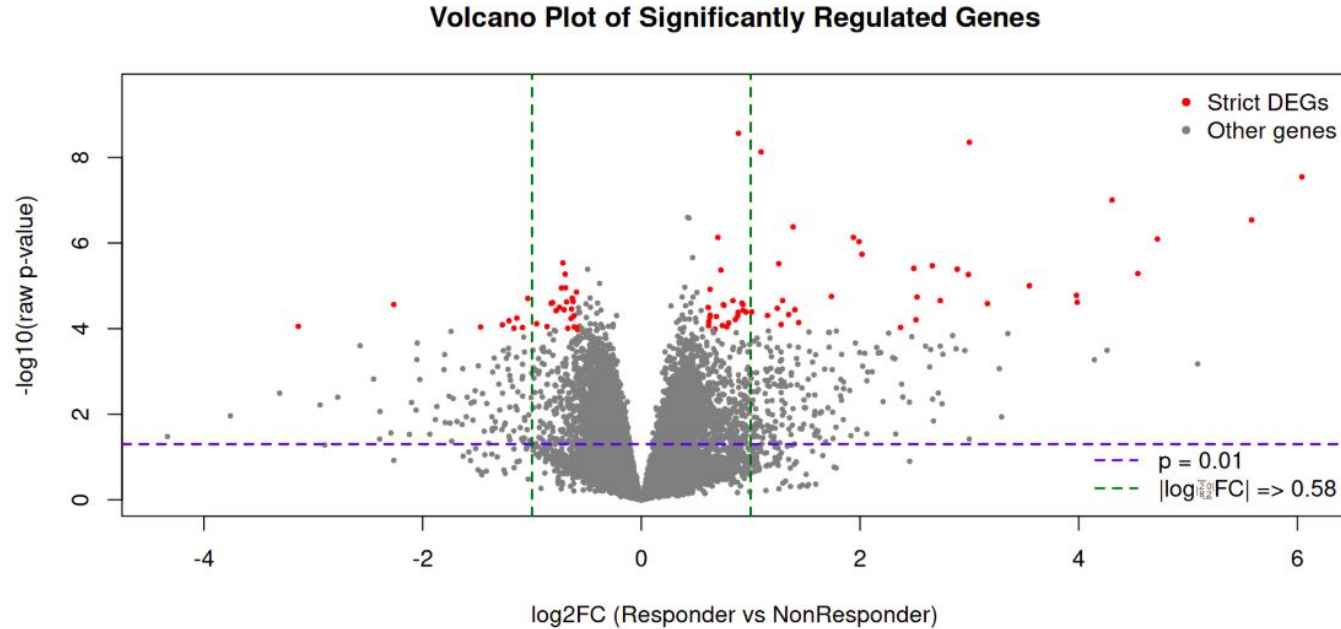


Figure 11: **Volcano plot displaying $\log_2$ fold changes against $-\log_{10}$ raw p-values.** Red points mark strict DEGs ($\text{FDR} < 0.01$ and $|\log_2\text{FC}| \geq 0.58$), while gray points represent other genes. Dashed lines indicate thresholds for fold change and p-value significance.

DEGs Significantly regulated expression genes

Gene ID	Biotype	Symbol	log ₂ FC	Avg log ₂ CPM	t- statistic
170482	protein_coding	CLEC4C	6.041	-0.707	7.531
166614	protein_coding	DCLK2	3.986	3.376	5.027
22982	protein_coding	DIP2C	2.888	3.101	5.672
1756	protein_coding	DMD	2.991	5.134	5.569
55567	protein_coding	DNAH3	-3.138	-0.146	-4.558
1840	protein_coding	DTX1	2.512	4.432	4.685
3759	protein_coding	KCNJ2	2.661	0.125	5.740
401944	protein_coding	LDLRAD2	3.548	-0.060	5.349
23547	protein_coding	LILRA4	3.980	3.752	5.160
643529	lincRNA	LINC00865	2.522	0.835	5.126
4023	protein_coding	LPL	4.720	2.876	6.268

Table 1: Top 20 differentially expressed genes meeting strict criteria (FDR < 0.01 and |log₂FC| ≥ 0.58), ranked by absolute fold change and sorted alphabetically by gene symbol.

Key observations include strong upregulation of immune related receptors (CLEC4C) and lipid metabolism genes (LPL) in responders, which may amplify drug sensitivity by altering both microenvironment signals and metabolic dependencies.

The **log₂ fold change (log₂FC)** indicates the magnitude of that difference: a positive value means the gene is expressed at higher levels in cells that die from the treatment, while a negative value means it is more abundant in resistant cells.

Gene Ontology functional analysis

Test the enrichment of DE genes in each of the thousands of gene sets defined by a GO term

GOstats:

1. Build a parameter object with information specifying the gene universe, the set of DE genes, the annotation package to use, etc.
2. Run the functional enrichment analysis.
3. Store and visualize the results.

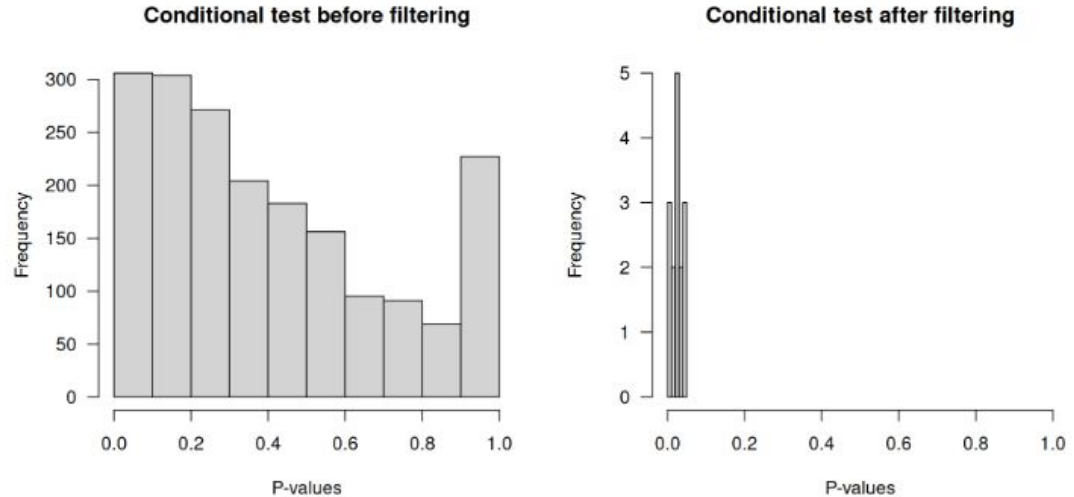


Figure 12: GEO functional enrichment results summarizing biological pathways and processes associated with differentially expressed genes

Filtering:

- Minimum value on the count & size (3 - 300)
- Order by odds-ratio

Term	Genes
cellular response to retinoic acid	LTK , TESC, PTK6
motile cilium assembly	DMD, TTC12, ARMC2
T cell mediated immunity	BTN3A3, SLAMF1, TNFRSF1B
microtubule bundle formation	TTC12, SPAG1, ARMC2
odontogenesis	LAMA5, ANKH, ROGDI
actin filament-based movement	PDPN, KCNJ2, EPB41L5
positive regulation of cation transmembrane transport	DMD, KCNJ2, TESC
epidermal growth factor receptor signaling pathway	LTK , PTK6 , REPS2
positive regulation of protein catabolic process	RNF41, CREBFB, TNFRSF1B

- Enrichment analysis highlights **immune-related** and **signaling pathways**.
- Key processes include:
 - **T cell–mediated immunity**
 - **Epidermal growth factor receptor (EGFR) signaling**
 - **Retinoic acid response**
- Several genes appear in **multiple pathways** (e.g., *PTK6*, *TNFRSF1B*).
- Suggests that **dasatinib sensitivity** may involve modulation of immune function and growth signaling.

Conclusion

- Our analysis revealed an 88 gene signature that clearly separates dasatinib-sensitive CLL cells by showing their weaker B-cell receptor signaling, a shift toward proapoptotic gene expression, and diminished reliance on microenvironmental survival signals.
- Gene Ontology enrichment pointed to immune-related processes such as T-cell immunity and EGFR signaling demonstrating how dasatinib's tyrosine kinase inhibition rewires both immune and growth factor pathways.
- The take home message is that assessing this concise gene set before treatment could help predict patient response to dasatinib and inform rational combination therapies.

Thank you for your attention

Blätte, Tamara J. et al. 2021. "Gene Expression Profiling Predicts Sensitivity of Chronic Lymphocytic Leukemia Cells to Dasatinib." HemaSphere 5 (1): e514.
<https://doi.org/10.1097/HS9.0000000000000514>.